

Not Alone, Not Forgotten:

Protective Role of Quantity and Quality of Social Network on Cognitive Decline

Chaeyeon Kim

Abstract

This study investigates how the quantity and quality of social connections jointly influence cognitive aging among older adults in South Korea. Using longitudinal data from the Korean Social Life, Health, and Aging Project (KSHAP), this paper examined whether outward-directed social connections—quantified by weighted out-degree centrality—predict changes in cognitive function measured by Mini-Mental State Examination (MMSE) scores over time. The weighted out-degree captures not only the number of social ties but also their emotional closeness, thereby including both the quantitative and qualitative dimensions of social networks. Fixed-effects and correlated random-effects panel models revealed that older adults with both larger and emotionally closer networks exhibited significantly less cognitive decline, even after controlling for depressive symptoms, self-rated health, income, sex, marital status, smoking, drinking, age, and education. The results suggest that having more close social networks serves as a protective factor against cognitive deterioration. This study highlights the importance of maintaining broad and emotionally engaging social networks as a low-cost, non-pharmacological strategy for promoting cognitive resilience in aging populations.

Keywords: cognitive aging, social network, quantity and quality, weighted out-degree, longitudinal analysis

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Cognitive aging poses a significant public health challenge in aging societies. While cognitive decline, including processing speed, episodic memory, and executive functions, is a natural path in cognitive aging (Craik & Salthouse, 2007; Small, Dixon, & McArdle, 2011; Lamar, Resnick, & Zonderman, 2003), it is still a serious issue in aging societies. While there are many factors, including smoking, poor diet, levels of physical activity, cognitive stimulation, and social relationships, that affect cognitive health (Craik & Salthouse, 2007; Small, Dixon, & McArdle, 2011; Lamar, Resnick, & Zonderman, 2003), many studies are focusing on the lifestyle pattern, including social networks, social support, and social relationships. While pharmacological treatments remain limited, recent research highlights the protective role of emotionally meaningful social networks in preserving cognitive health.

Previous research has shown that those in plentiful social networks are associated with better cognitive functioning and less cognitive decline. Larger and more active social networks, according to Holtzman et al. (2004), provide cognitive stimulation and are associated with slower cognitive decline. Beyond network size, Seeman et al. (2001) emphasized that relationship quality—particularly emotional support—plays a more influential role in preserving cognitive function, while relational strain can have detrimental effects. In addition, recent work by Roh et al. (2022) further demonstrated that social support can buffer the negative effects of depressive symptoms on cognitive outcomes.

Numerous studies have investigated the correlation between social relationships and cognitive outcomes; however, few have methodically examined the quality of social networks. Prior research often represented social networks only based on quantitative features—such as the number of friends, relatives, or acquaintances—while overlooking the

qualitative strength of those ties (Glei et al., 2005; Hsu, 2007; Ho et al., 2001). Yet, the frequency of contact and emotional closeness within one's network may play a crucial role in maintaining cognitive functioning. As Granovetter (1973) argued, the strength of a tie involves not only its existence but also its frequency, emotional intensity, and reciprocity—dimensions that determine whether a relationship can provide meaningful support and cognitive stimulation in daily life.

To address this limitation, the present study adopts a *weighted out-degree* measure that quantifies both the closeness and quantity of social networks. In this context, the size of the discussion network refers to the number of people with whom the respondent discusses important or serious matters, while the weight of each tie reflects how emotionally close the respondent feels to that person. Previous research on weighted networks has demonstrated that nodes vary not only in the quantity of alters to which they are connected but also in the intensity of their relational maintenance. Neglecting tie weights can lead to a misrepresentation of the true extent of social engagement or influence (Opsahl, Agneessens, & Skvoretz, 2010). Following this measure, the weighted out-degree was used to capture both the frequency and depth of interaction because ties with higher weights are likely to exert greater social and cognitive influence than occasional contacts. Utilizing the KSHAP data, which record the frequency and perceived strength of respondents' outgoing contacts, the *weighted out-degree* offers a methodologically robust indicator of social networks.

Additionally, some previous studies only used five questions to evaluate cognitive function. In contrast, this paper used the Korean version of the MMSE, which includes 20 items, to provide a more comprehensive and accurate assessment of older adults' cognitive ability. Furthermore, relatively few studies have examined these associations in Asian—particularly Korean—populations, as most existing research has been conducted in Western contexts.

While numerous studies have found an association between cognitive function and social networking, much of this research has employed cross-sectional data. Thus, the association may be confounded by time-invariant characteristics, including personality, early-life experiences, or genetic predispositions. (Cacioppo & Hawkley, 2009; Kuiper et al., 2016). To address this, the present study employs panel data using fixed-effects (FE) and correlated random-effects (CRE) models and provides stronger evidence that having a social network itself is protective against cognitive decline. The fixed effect model accounts for all unobserved time-invariant traits by examining within-person changes in social relationships and cognitive function. Thus, this approach helps eliminate confounding effects due to stable individual characteristics such as personality, education, and early-life conditions. The correlated random effects models further estimate both within- and between-person effects. (Mundlak, 1978; Wooldridge, 2018), explaining well how both short-term and long-term changes in social networks are related to cognitive functioning. By combining these methods, this study provides a more statistically rigorous test of how changes in social networks and closeness within them are associated with cognitive functioning.

Another limitation of previous research is that the results were not coherent. The majority of research showed a significant association between social network size and frequency of contact with measures of global cognition (Barnes et al., 2004; Holtzman et al., 2004; Béland et al., 2005; Kimura et al., 2017; Zunzunegui et al., 2003; Ho et al., 2001). *Seeman et al. (2001)* and *Glei et al. (2005)* reported no association between social networks and global cognition. *Hughes et al. (2008)* found no association between social network size and measures of episodic memory, attention, processing speed, or global cognition.

To address these gaps, this study uses data from the first wave of the Korean Social Life, Health, and Aging Project (KSHAP) to examine how the quantity and quality of social networks are effective in cognitive function in later life.

Method

Data

This study uses data from the first wave of the Korean Social Life, Health, and Aging Project (KSHAP). The survey was conducted in 2011 and 2013 and targeted all individuals aged 60 and over, along with their spouses (regardless of age), residing in a village (the smallest administrative unit) in Township K, Ganghwa Island, South Korea. A total of 814 respondents participated in the first wave and 590 in the third wave. Among them, 571 individuals who could be identified as the same respondents across both waves were included in the final analysis.

Measures

Covariates: In all regression models, age, sex, education level, income, marital status, Instrumental Activities of Daily Living, and depressive symptoms at wave 1 were controlled.

Depressive symptoms: Depressive symptoms were assessed using 20 items from the Korean CES-D scale, with three positively worded items reverse coded and all items rescaled from 1–4 to 0–3. The total score ranged from 0 to 60, with higher scores indicating more severe symptoms.

Weighted Out-Degree Centrality:

Respondents were first asked to identify people with whom they had discussed important or serious matters over the past year:

“People often discuss their important issues with others. For example, it may be a good thing or a bad thing that personally happened to you, or it could be a concern or interest you are having. Over the past year, how many of these people have you had?”

After listing all discussion partners, respondents were then asked to indicate their perceived intimacy with each person using the question:

“How close do you think you are to [name]?” with response options ranging from *“not very close,” “somewhat close,”* and *“very close”* to *“extremely close.”*

The weighted out-degree for each respondent was computed by summing the number of discussion partners, weighted by the reported level of intimacy. This metric captures both the size and depth of social relationships.

Discussion networks were widely used to examine the social network in many previous studies. (Peter V. Marsden, 1987; Oxana Mikhaylova & Sofia Dokuka, 2025) Using discussion networks, the quality and quantity of social networks that actually affected the respondents over the past year could be captured.

Cognitive Measure: Cognitive change was computed by using total MMSE scores (0–30). In Wave 1, "Not measured" was treated as "Incorrect," whereas in Wave 3, "Not measured" was treated as "Don't know." In Wave 1, responses marked as “Not measured” were treated as incorrect responses and included as such in the analysis. In contrast, in Wave 3, “Not measured” responses were excluded from the analysis by coding them as missing (NA).

Statistical Methods: The random-effect (RE) model, fixed-effects (FE), and correlated random-effects (CRE) models were used to analyze longitudinal changes in cognitive function. The fixed effect model controls for unobserved time-invariant individual heterogeneity by focusing on within-person variation over time (Hsiao, 2014; Baltagi, 2008).

Compared to the fixed effect model, the RE model assumes that individual-specific effects are uncorrelated with explanatory variables—a strong assumption often violated in social data. It was used to compare the results of the fixed-effect model in this paper. To further complement the fixed effect model, the CRE model (Mundlak, 1978; Wooldridge, 2018) allows decompositions of effects into within-person and between-person components, helping reduce the endogeneity issues. Thus, the sole effect of having social networks on cognitive function can be identified.

Results

Table 1

Frequency of Categorical Variables

Variable	Category	Count (n = 571)	Percent (%)
Education	No formal education	164 / 197	29.2 / 34.5
	Elementary	245 / 223	43.6 / 39.1
	Middle School	79 / 76	14.1 / 13.3
	High School	54 / 52	9.6 / 3.1
	College or higher	20 / 23	3.6 / 4.0
Income Level	Low	292 / 311	62.1 / 59.4
	Mid-low	115 / 122	24.5 / 23.3
	Mid	48 / 50	10.2 / 9.5
	High	15 / 41	3.2 / 7.8
Sex	Male	234 / 234	41.0 / 41.0
	Female	337 / 337	59.0 / 59.0
Smoker	Smoker	67 / 58	11.7 / 10.2
	Non-smoker	504 / 513	88.3 / 89.8
Spouse Status	No Spouse	126 / 142	22.1 / 24.9
	Has Spouse	445 / 429	77.9 / 75.1

Note. Counts and percentages represent participants who completed both Wave 1 and Wave 3 of the survey. Values are presented as Wave 1/ Wave 3. Participants who dropped out between waves were excluded from this table.

Table 2

Descriptive Statistics of Continuous Variables

Variable	Wave 1	Wave 3
MMSE	24.28 (4.33)	23.50 (5.49)
log(Weighted Out-degree)	2.38 (0.48)	2.34 (0.71)
CES-D	10.90 (7.75)	7.32 (8.61)
Self-rated health	2.64 (0.79)	3.38 (1.05)
Age	70.80 (7.14)	72.93 (7.13)
Drinking	0.72 (1.59)	0.78 (1.90)

Note. Mean (SD). Only participants with data at both Wave 1 and Wave 3 are included.

Table 1 presents the frequency distributions of key categorical variables. Table 2 summarizes the descriptive statistics of the continuous variables. The panel consists of 571 participants who participated in both the survey in wave 1 and wave 3, while the total number of participants at wave 1 was 814. 243 participants dropped out. There was a slight shift in education levels and smokers. The figure shows a slight improvement in income levels overall at wave 3, with the share of the high-income group increasing from 3.2% to 7.8%, particularly among participants who do not have a spouse. Sex composition remained constant across waves (41% male, 59% female).

Table 2 presents the descriptive statistics for key continuous variables. The average Mini-Mental State Examination (MMSE) score decreased from 24.28 (SD = 4.33) in Wave 1

to 23.50 (SD = 5.49) out of 30 in Wave 3, indicating a slight cognitive deterioration over time. The mean value of the log-transformed out-degree slightly decreased from 2.38 (SD = 0.48) to 2.34 (SD = 0.71), and the minimum and maximum values on both waves were 0 and 28. In contrast, depressive symptoms measured by the CES-D showed a moderate improvement, decreasing from 10.90 (SD = 7.75) to 7.32 (SD = 8.61). The minimum CESD score was 0 in both waves and 51 and 60 in wave 1 and wave 3, respectively. Self-rated health ranges from 1 to 5, and it improved from 2.64 (SD = 0.79) to 3.38 (SD = 1.05), while the mean age increased from 70.80 (SD = 7.14) to 72.93 (SD = 7.13), reflecting the longitudinal nature of the data. The average level of drinking behavior remained relatively stable, decreasing from 0.72 to 0.78 times per week. Females and males accounted for 59% and 41% of the sample in both waves, respectively.

Table 3

Correlation Among Depression, Network, and Cognitive Function Variables

Variable	MMSE (Wave 1)	MMSE (Wave 3)	Weighted Out-Degree (Wave 1)	Weighted Out-Degree (Wave 3)
MMSE (Wave 1)	1			
MMSE (Wave 3)	0.512	1		
Weighted Out-Degree (Wave 1)	0.157	0.167	1	
Weighted Out-Degree (Wave 3)	0.232	0.303	0.198	1

Note. Values are Pearson's r using pairwise complete observations.

Correlation analysis showed individuals with larger and intimate social networks at baseline tended to have higher cognitive function at both wave 1 and wave 3 ($r = 0.157$, 0.167). Larger and intimate social networks at wave 3 also had similar associations with cognitive function ($r = 0.233$, 0.304). The association between cognitive function at waves 1 and 3 was 0.512 , indicating that there has been a moderate change in cognitive function itself across the time. In contrast, the correlation between weighted out-degrees was relatively low ($r = 0.20$), indicating substantial changes over time ($r = 0.20$). This high within-person variation in social network support using panel models such as the fixed model and CRE model accurately captures within-person changes over time in social networks and better explains the effect of social networks on cognitive function. Cognitively higher-functioning individuals were more likely to sustain or expand their social networks over time, considering the correlation between MMSE score at w1 and weighted out-degree at wave 3 ($r = 0.23$).

Table 4

Table 4. Fixed-Effect, Random-Effect, and Correlated Random-Effect Models Predicting Cognitive Function

	Fixed Effect	Random Effect	CRE
Intercept		3.470 (0.126)***	2.780 (0.069)***
Weighted Out-degree	0.052 (-0.021)*	0.067 (0.014)***	0.047 (0.019)*
Weighted Out-degree (Mean)			0.058 (0.028)*
CES-D	-0.003 (0.002)*	-0.002 (0.001)*	-0.002 (0.001)
CES-D (Mean)			-0.001 (0.002)
Self-Rated Health	-0.008 (0.011)	-0.003 (0.008)*	-0.018 (0.009)+
Self-Rated Health (Mean)			0.043 (0.018)*

Other Control Variables	✓	✓	✓
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Note. The dependent variable is the residualized Mini-Mental State Examination (MMSE) score. Coefficients are presented with standard errors in parentheses. $p < .05$, $\mathbf{p} < .01$, $\mathbf{\mathbf{p}} < .001$, $+p < .10$. All models control for age, sex, education, income, smoking, drinking, and spouse status.

While all models control for depressive symptoms, self-rated health, income, sex, marital status, smoking, drinking, age, and education, the results of the fixed-effect model suggested that those with higher weighted out-degrees were associated with higher MMSE scores, indicating that individuals with larger and more intimate social networks tended to have better cognitive functioning over time (0.052, $p = 0.031$). Higher depressive symptoms were associated with greater cognitive decline (-0.003 , $p = 0.042$). Age was a significant negative predictor of cognitive function (-0.026 , $p = 0.002$). Those who achieved higher education had better cognitive performance as follows: elementary education (0.130, $p < 0.001$), middle school (0.197, $p < 0.001$), high school (0.198, $p = 0.007$), and college or higher (0.209, $p = 0.122$). Individuals who drank moderately also exhibited better cognitive performance (0.018, $p = 0.049$). Compared to the low-income group, individuals in the middle-income and high-income groups demonstrated significantly higher cognitive functioning scores (0.115, $p < .01$; 0.102, $p < .05$, respectively). This indicates that higher income levels may provide greater access to resources or environments that promote cognitive health. No significant difference was observed between the mid-low- and low-income groups. The overall coefficient of determination (R^2) was 0.162.

The Hausman specification test indicated that the fixed-effects model was statistically more suitable for this analysis ($\chi^2 = 42.999$, $df = 13$, $p = 4.49e-05$), suggesting that unobserved heterogeneity was correlated with explanatory variables.

Although the Hausman test supported the fixed model, the random-effects model produced consistent patterns. A higher weighted out-degree remained significantly associated with cognitive function (0.067, $p < 0.001$). Income was positively related to cognitive performance (0.034, $p = 0.018$), as was higher education (college or higher: 0.122, $p = 0.013$). Similar to the fixed model, higher age (-0.007 , $p < 0.001$) and depressive symptoms (-0.002 , $p = 0.021$) were associated with lower cognitive scores. The random-effects model explained 24.4% of the variance ($R^2 = 0.244$).

In the correlated random-effects (CRE) model, both within- and between-person variations of social network out-degree were positively related to cognitive function. People with bigger and closer social networks did better on cognitive tests (0.047, $p = 0.019$; 0.058, $p = 0.027$). Between-person differences in self-rated health were positively associated with MMSE score (0.043, $p = 0.016$), while within-person changes showed a marginal negative association (-0.018 , $p = 0.068$). Because self-rated health is based on how a person feels about their health rather than their actual medical condition, this pattern may indicate that people who are having cognitive problems are overestimating their overall health. Education level remained a strong positive predictor, consistent across models: elementary (0.195, $p < 0.001$), middle school (0.252, $p < 0.001$), high school (0.249, $p < 0.001$), and college or higher (0.180, $p < 0.001$). Increases in income (0.034, $p = 0.025$) were related to cognitive outcomes, indicating that economic improvements benefit cognitive health despite a slight negative cross-sectional association. Higher mean income (0.039, $p = 0.039$) was associated with higher MMSE scores. Age was negatively associated with cognitive outcomes, as expected (-0.007 , $p < 0.001$). The CRE model explained 23.8% of the variance ($R^2 = 0.238$).

This study investigates how the quantity and quality of social networks influence cognitive functioning among older adults in Korea. Cognitive decline and dementia are significant

public health issues, and it is increasingly vital to promptly identify cognitive decline prior to a clinical diagnosis. By focusing on MMSE-based cognitive function scores rather than dementia onset, this study captures earlier and more subtle variations in cognitive aging. While dementia is a major public health concern, cognitive decline begins long before clinical diagnosis. By focusing on MMSE-based cognitive function scores rather than dementia onset, this study captures earlier and more subtle variations in cognitive aging that are socially determined.

Discussion

Consistent with prior research, this study found that both the size and emotional closeness of social networks were positively associated with cognitive functioning (Seeman et al., 2001; Fratiglioni et al., 2004; Kuiper et al., 2016). Drawing on the social integration theory, individuals embedded in broader and more interactive networks could highly engage in social activities, receive more emotional support and interpersonal advice, and benefit from social monitoring that discourages risky behavior—for example, being told to exercise or quit smoking by other people—compared to those with limited social networks. Moreover, consistent with the cognitive reserve hypothesis (Stern, 2002), frequent and meaningful social interactions may serve as cognitively stimulating activities that help protect individuals from neural deterioration, as these interactions require communication and attention. Older adults with both larger and emotionally closer networks have more opportunities to attend to others, comprehend language, and recall shared experiences. These interactions can simulate complex cognitive operations and help protect against cognitive decline. Another possible explanation is that emotional and instrumental support could decrease both physiological and psychological consequences of stress. Frequent exposure to stress can damage hippocampal neurons by releasing cortisol and disrupt prefrontal cortical function, which accelerates

cognitive decline. (Lupien et al., 2009). Since a close network can provide emotional support that relieves stress and offers more emotionally uplifting experiences, social support within this network could help maintain cognitive health.

Methodologically, the present study improves on prior work by focusing on quantitative aspects of social networks. Previous research tended to measure the social network simply by the numbers of known individuals, including friends, families, and neighbors. (Glei, 2005; Hsu, 2007; Ho, 2001). While social support has been complementarily used to represent the qualitative aspects of social networks, it does not capture how close respondents actually feel toward each person with whom they have meaningful connections. Unlike surveys that ask generally about support from “family or friends,” the KSHAP survey used in this paper asked respondents to list the people with whom they had discussed important or serious matters over the past year. These discussion partners can be emotionally close kin or friends but also co-workers, neighbors, or other contacts with whom respondents can talk about personal events, problems, or concerns. After identifying these alters, respondents were additionally asked how close they felt to each person. We used this reported closeness as a weight and constructed a weighted out-degree index. This two-step procedure allows us to distinguish between (a) relationships that are merely present in someone’s social environment and (b) relationships that are both activated for important discussions and perceived as closer.

This approach is preferable to simple counts of relatives or friends because previous studies could not tell whether listed network members were actually contacted or influential in everyday decision-making (Glei, 2005; Hsu, 2007; Ho, 2001). By contrast, the present measure is based on enacted, issue-driven interactions and then refined by the respondent’s subjective closeness, providing a more realistic indicator of the network that may affect cognitive functioning. In line with Opsahl, Agneessens, and Skvoretz (2010), weighting ties in this way helps avoid treating all relationships as equally important and reflects that some

people interact more often or have stronger influence than others. This measure follows Granovetter's (1973) view that strength of ties consists of frequency, emotional intensity, and reciprocity, which should not be understood merely in terms of the presence or absence of a relationship.

Limitation

The weak and inconsistent association between self-rated health and cognitive performance may reflect the subjective nature of self-assessed health among older adults. As cognitive impairment progresses, individuals may overestimate their health status, leading to potential measurement bias. While self-rated health was the only variable that can represent the physical ability of the respondents, future studies should include more comprehensive and accurate measures to better capture the physical health condition in later studies. Despite the use of two panels in the present study, potential confounders may still exist. Although the CRE model partially accounts for unobserved heterogeneity by separating within- and between-person effects, it cannot fully eliminate bias from time-varying unmeasured factors. Moreover, social and environmental variables—such as neighborhood cohesion, access to community resources, or caregiving responsibilities—were not included, which might have influenced both the quantity and quality of social interactions. Future research should address these omitted variables by incorporating richer behavioral and contextual data or using longer periods of longitudinal data to reduce the reversed causality. Moreover, While the weighted out-degree used in this paper considered the closeness within the social network, it did not fully capture the perceived levels of intimacy and reciprocity. Future studies should refine network quality, covering both emotional and functional dimensions of ties.

Conclusion

Cognitive decline and dementia have emerged as increasingly critical concerns among aging populations. Despite the development and effectiveness of numerous medical and pharmacological interventions after dementia onset, they failed to prevent early cognitive decline. Thus, it is essential to identify social factors that can alleviate cognitive decline.

The present study advances this line of research by emphasizing the protective function of social networks—both in terms of quantity and quality—against cognitive decline in older adults. Compared to previous studies that focused only on the quantity of social relationships or failed to capture the degree of closeness toward each individual, our study measured both the number of discussion partners and the level of emotional closeness with each of them. By considering the closeness between each individual in the discussion network, the results suggest that having larger and stronger networks can be protective against cognitive decline.

This study situates itself at the intersection of social network and cognitive aging research by arguing that those immersed in larger and emotionally close networks have better cognitive outcomes. The results indicate that fostering numerous emotionally intimate relationships may serve as a cost-effective, non-pharmaceutical strategy to promote healthy cognitive aging in later life. It can contribute to social policies aimed at fostering more interconnected and cognitively resilient communities.

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